

# RADHEY KALRA

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Building ML/audio systems and self-hosted infrastructure, with a current focus on **audio denoising**, cloud platforms, and scalable developer tooling.

## EDUCATION

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### Bachelor of Technology in CSE

*Model Institute of Engineering and Technology*

Aug. 2023 – May 2027

*Jammu and Kashmir, India*

## TECHNICAL SKILLS

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**Languages:** Python, C/C++, TypeScript, Go, Rust, SQL, Bash, QML, LaTeX

**Frameworks & Platforms:** SvelteKit, React, Next.js, Astro, Tailwind CSS, React Native, Expo, Bun, Node.js

**Developer Tools:** Git, Docker, CMake, CI/CD, Cloudflare, Vercel, Kubernetes, Terraform

**Databases & Cloud:** Convex, PostgreSQL, MySQL, AWS, [GCP](#)

**Libraries:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Manim

**Domains:** Deep Learning, Audio Processing, Linux Servers, Full-Stack Development, Networking

## EXPERIENCE

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### Cloud Lead

*GDGoC (Google Developer Groups on Campus), MIET Jammu*

Aug. 2025 – Present

*Jammu and Kashmir, India*

- Lead cloud learning initiatives by organizing hands-on workshops and guided sessions around **Google Cloud** services and developer tooling.
- Build and maintain a **tutorial platform** ([Live/GitHub](#)) containing lab walkthroughs, notes, and video resources to help students complete Study Jam skill badges efficiently.

### Cloud and DevOps Lead

*AWS Cloud Club MIET Jammu*

Jan. 2026 – Present

*Jammu and Kashmir, India*

- Lead cloud and DevOps initiatives as part of the core team by mentoring students through workshops, collaborative projects, and technical sessions.
- Promote adoption of **AWS**, containerization, and infrastructure best practices across club activities with focus on deployment, automation, and scalability.

## PROJECTS

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### U-Net Audio Filter | *Audio Enhancement by removing background noise* | [GitHub](#)

- Current research project on audio enhancement using a **custom U-Net-inspired architecture** and end-to-end training pipeline for denoising and source separation.
- Developing the workflow with **Python, PyTorch, and audio DSP preprocessing**, including custom data handling, spectrogram-based modeling, and evaluation tooling.
- In development, but already outperforming existing **open-source baselines on select metrics**, showing promising near-SOTA results in early experiments.

### Homelab File Manager | *Self-hosted file management system for Linux servers* | [Live](#) | [GitHub](#)

- Built a **fast, lightweight** file manager for Linux servers using **Svelte** and **Go**, supporting multi-mount browsing, chunked transfers, and real-time file updates.
- Added support for **most media content** with live preview/streaming, plus **Monaco Editor** integration for viewing and editing text-based files directly in the browser.

### Homelab Infrastructure | *Self-hosted server for storage, remote access, and service deployment* | [GitHub](#)

- Built a personal homelab on repurposed hardware running **Ubuntu Server**, using **Docker Compose** for service orchestration, **Cloudflare Tunnel** for secure remote access, and **Traefik** for reverse proxying.
- This is where I host tools such as **ViDown** ([Live](#) / [GitHub](#)) and **AnyConverter** ([Live](#) / [GitHub](#)), which I use directly in my day-to-day workflow.
- Maintained a custom **CI/CD workflow** that tracks GitHub changes, pulls updates onto the server, syncs with self-hosted **Gitea**, and automatically redeploys services.

### HowTo | *Minimal technical tutorials and guides site* | [Live](#) | [GitHub](#)

- Built a technical tutorials website using **Astro**, **Tailwind CSS**, and **Bun**, with structured content collections, syntax highlighting, and a clean reading experience.
- Actively publish technical guides across both **hardware** and **software** topics, with practical documentation.

#### Community Leaderboard | *Dashboard for event tracking, rankings, and community management* | [Live](#) | [GitHub](#)

- Developed a community management dashboard with **Next.js**, **Convex**, and **Clerk** for user profiles, event tracking, points allocation, and leaderboard views.
- Designed admin-focused workflows for managing events, monitoring participation, and maintaining rankings through clean, responsive interfaces.

#### Hotel Booking | *Full-stack booking platform with auth and admin workflows* | [GitHub](#)

- Built a hotel booking application using **Svelte**, **Node.js**, **Express**, and **MySQL**, featuring authentication, booking management, and admin-facing controls for handling reservations.

#### Terminal Doom | *Doom-inspired terminal game built with ncurses* | [GitHub](#)

- Created a Doom-styled game in the terminal using **C++**, **ncurses**, and **CMake**, translating arcade-style gameplay and rendering into a text-based interface.

#### Pico32 | *Serial-only TUI for ESP32 Marauder on generic ESP32 hardware* | [Live](#) | [GitHub](#)

- Developed a lightweight terminal interface for ESP32 Marauder using **C++** and PlatformIO, enabling Wi-Fi and Bluetooth testing workflows over serial on low-cost ESP32 boards.
- Focused on portability and usability by adapting a hardware security toolchain into a compact, keyboard-driven TUI experience.

#### dots-niri | *Automated Arch Linux dotfiles and installer for the Niri compositor* | [GitHub](#)

- Built a personal **Niri** rice and automation setup with an installer that provisions packages, symlinks configs, copies assets, and enables desktop services on Arch Linux.
- Extended the same Linux desktop workflow with companion tools including **WallpyGui** ([GitHub](#)), a wallpaper manager for Wayland compositors, and **iwd-applet** ([GitHub](#)), a tray applet for menu-driven wireless management through **iwd**.

#### TL;DR AI | *Desktop app to summarize WhatsApp chats using LLMs* | [GitHub](#)

- Built an **Electron** desktop application with **React** and **TypeScript** to summarize WhatsApp group chats using OpenRouter-compatible AI models.
- Users log in via QR and enter their OpenRouter API key.
- Designed the workflow so users can process their own chat data with flexible model choices, making the app practical for personal productivity use cases.

#### Tenant Manager | *Mobile app for tracking tenants and electricity usage* | [GitHub](#)

- Built a mobile management app with **React Native**, **Expo**, and **TypeScript** to maintain tenant records and log electricity usage in a simpler field-friendly workflow.

## CERTIFICATIONS & PROGRAMS

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Google Cybersecurity | [Certificate](#)

Cisco Cybersecurity Operations Fundamentals | [Certificate](#)

IBM SkillsBuild: AI Agent Architect | [Certificates](#)

IDE Bootcamp (AICTE / MoE) | [Certificate](#)

## LEADERSHIP & ACTIVITIES

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### Team Leader

*Internal Hackathon & Smart India Hackathon (SIH 2024 & 2025)*

*MIET, Jammu*

- Led multidisciplinary teams for the college's internal hackathon and Smart India Hackathon selection rounds in both **2024** and **2025**.
- Reached the **top 9 teams in the college** in both years by driving collaborative problem-solving, rapid prototyping, and solution design for real-world challenges.

### Team Leader

*Innovation, Design, and Entrepreneurship (IDE) Bootcamp*

*Feb. 2025*

*IIT Jammu*

- Participated in a 5-day immersive bootcamp organized by IIT Jammu in collaboration with the Wadhwani Foundation, AICTE, and MoE's Innovation Cell.
- Led a team in developing a “Sports Match Finder” app, applying principles of design thinking and the Lean Canvas model to address market needs.
- Engaged in hands-on activities and real-world case studies, gaining insights into business modeling, product design, and entrepreneurial strategies.